

GraphTheorySymbol:

A set of $\text{\LaTeX}$ commands to have symbols designed for
Graph Theory

Laurent Frédéric Bernard François Lyaudet
Website:<https://lyaudet.eu/laurent/>
Email:Laurent.Lyaudet@gmail.com

v1.4.0 2026/09/12

Abstract

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

Contents

1	Introduction	1
2	Usage	2
2.1	Memo or for the impatient	2
2.2	Detailed usage	4
2.3	Good practices	15
3	Implementation	15
4	Acknowledgments	27
5	Support	27

1 Introduction

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com:LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSOI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSOIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$

Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADMR/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSOIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		$u \overset{\circ}{\sim} v$
\GTSadjacencyRelation/	\GTSaR/		$u \overset{\circ}{\sim} v$
\GTSunadjacency/	\GTSu/		$u \overset{\circ}{\sim} v$
\GTSunadjacencyRelation/	\GTSuR/		$u \overset{\circ}{\sim} v$
\GTSdirectedAdjacencyUp/	\GTSdAU/		$u \overset{\circ}{\sim} v$
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		$u \overset{\circ}{\sim} v$
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		$u \overset{\circ}{\sim} v$
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		$u \overset{\circ}{\sim} v$
\GTSdirectedAdjacencyDown/	\GTSdAD/		$u \overset{\circ}{\sim} v$
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		$u \overset{\circ}{\sim} v$
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		$u \overset{\circ}{\sim} v$
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		$u \overset{\circ}{\sim} v$
\GTSincidence/	\GTSi/		$v \overset{\circ}{\sim} e$
\GTSincidenceRelation/	\GTSiR/		$v \overset{\circ}{\sim} e$
\GTSnegatedIncidence/	\GTSnI/		$v \overset{\circ}{\sim} e$
\GTSnegatedIncidenceRelation/	\GTSnIR/		$v \overset{\circ}{\sim} e$
\GTSoutIncidence/	\GTSOI/		$v \overset{\circ}{\sim} e$
\GTSoutIncidenceRelation/	\GTSOIR/		$v \overset{\circ}{\sim} e$
\GTSnegatedOutIncidence/	\GTSnOI/		$v \overset{\circ}{\sim} e$
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		$v \overset{\circ}{\sim} e$
\GTSinIncidence/	\GTSiI/		$v \overset{\circ}{\sim} e$
\GTSinIncidenceRelation/	\GTSiIR/		$v \overset{\circ}{\sim} e$
\GTSnegatedInIncidence/	\GTSnII/		$v \overset{\circ}{\sim} e$
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		$v \overset{\circ}{\sim} e$

Long name	Short name	Result	In context
\GTSmetaNot/	\GTSmN/	$\neg$	$\neg \overset{\circ}{\sim}$
\GTSmetaAnd/	\GTSmA/	$\wedge$	$\overset{\circ}{\sim} \wedge \overset{\circ}{\sim}$
\GTSmetaOr/	\GTSmO/	$\vee$	$\overset{\circ}{\sim} \vee \overset{\circ}{\sim}$

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Directed graph:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Oriented graph:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Prefix notation:

$\exists u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

$\exists v, e, v \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} e$.

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

`\GTS@newprotectedcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`{\langle relation \rangle}` Non-empty implies that relation spacing will be added.

`\GTSxa` Versatile macros for adjacency that can be given options.

`\GTSxadj` Argument:

`\GTSxadjacency` [`\langle optionsList \rangle`] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,

secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,

monochrome or mono or mon or m (equivalent to mC=black,sC=black),

negated or not or n,

up or u,

down or d,

relation or rel or r.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small

`\GTSadj` dots in the middle of the two circles corresponding to the graph vertices. But it was too

`\GTSa` similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property

to remind of the classical textual notation of adjacency relation as Adj(...).

A.\GTSadjacency/A.
or A.\GTSadj/A.
or A.\GTSa/A.
will yield $A \circ A$.
u \GTSadjacency/ v
or u \GTSadj/ v
or u \GTSa/ v
will yield $u \circ v$.
\$u \GTSadjacency/ v\$
or \$u \GTSadj/ v\$
or \$u \GTSa/ v\$
will yield $u \circ v$.

\GTSadjacencyRelation This macro will draw an adjacency symbol.

\GTSadjRel \$u\GTSadjacencyRelation/v\$
\GTSaR or \$u\GTSadjRel/v\$
or \$u\GTSaR/v\$
will yield $u \circ v$.

\GTSunadjacency This macro will draw an unadjacency symbol.

\GTSunadj A.\GTSunadjacency/A.
\GTSu or A.\GTSunadj/A.
or A.\GTSu/A.
will yield $A \oslash A$.
u \GTSunadjacency/ v
or u \GTSunadj/ v
or u \GTSu/ v
will yield $u \oslash v$.
\$u \GTSunadjacency/ v\$
or \$u \GTSunadj/ v\$
or \$u \GTSu/ v\$
will yield $u \oslash v$.

\GTSunadjacencyRelation This macro will draw an unadjacency symbol.

\GTSunadjRel \$u\GTSunadjacencyRelation/v\$
\GTSuR or \$u\GTSunadjRel/v\$
or \$u\GTSuR/v\$
will yield $u \oslash v$.

\GTSdirectedAdjacencyUp This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

\GTSdirAdjUp A.\GTSdirectedAdjacencyUp/A.
or A.\GTSdirAdjUp/A.
or A.\GTSdAU/A.
will yield $A \circ A$.
u \GTSdirectedAdjacencyUp/ v
or u \GTSdirAdjUp/ v
or u \GTSdAU/ v
will yield $u \circ v$.
\$u \GTSdirectedAdjacencyUp/ v\$
or \$u \GTSdirAdjUp/ v\$
or \$u \GTSdAU/ v\$
will yield $u \circ v$.

\GTSdirectedAdjacencyUp-Relation This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

\GTSdirAdjUpRel \$u\GTSdirectedAdjacencyUpRelation/v\$
\GTSdAUR or \$u\GTSdirAdjUpRel/v\$
or \$u\GTSdAUR/v\$
will yield $u \circ v$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol.

`AdjacencyUp` $A.\GTSnegatedDirectedAdjacencyUp/A.$

`\GTSnegDirAdjUp` or $A.\GTSnegDirAdjUp/A.$

`\GTSnDAU` or $A.\GTSnDAU/A.$

will yield $A.\overleftarrow{A}.$

$u \ \GTSnegatedDirectedAdjacencyUp/ \ v$

or $u \ \GTSnegDirAdjUp/ \ v$

or $u \ \GTSnDAU/ \ v$

will yield $u \overleftarrow{} v.$

$\$u \ \GTSnegatedDirectedAdjacencyUp/ \ v\$$

or $\$u \ \GTSnegDirAdjUp/ \ v\$$

or $\$u \ \GTSnDAU/ \ v\$$

will yield $u\overleftarrow{}v.$

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol with relation
`AdjacencyUpRelation` spacing in math mode.

`\GTSnegDirAdjUpRel` $\$u\GTSnegatedDirectedAdjacencyUpRelation/v\$$

`\GTSnDAUR` or $\$u\GTSnegDirAdjUpRel/v\$$

or $\$u\GTSnDAUR/v\$$

will yield $u \overleftarrow{} v.$

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left
`\GTSdirAdjDown` of the symbol should be considered the target of the arc; the operand on the right of the
`\GTSdAD` symbol should be considered the source of the arc.

$A.\GTSdirectedAdjacencyDown/A.$

or $A.\GTSdirAdjDown/A.$

or $A.\GTSnDAD/A.$

will yield $A.\overrightarrow{A}.$

$u \ \GTSdirectedAdjacencyDown/ \ v$

or $u \ \GTSdirAdjDown/ \ v$

or $u \ \GTSnDAD/ \ v$

will yield $u \overrightarrow{} v.$

$\$u \ \GTSdirectedAdjacencyDown/ \ v\$$

or $\$u \ \GTSdirAdjDown/ \ v\$$

or $\$u \ \GTSnDAD/ \ v\$$

will yield $u\overrightarrow{}v.$

`\GTSdirectedAdjacencyDown-` This macro will draw a directed adjacency downward symbol with relation spacing in
`Relation` math mode.

`\GTSdirAdjDownRel` $\$u\GTSdirectedAdjacencyDownRelation/v\$$

`\GTSdADR` or $\$u\GTSdirAdjDownRel/v\$$

or $\$u\GTSdADR/v\$$

will yield $u \overrightarrow{} v.$

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol.

`AdjacencyDown` $A.\GTSnegatedDirectedAdjacencyDown/A.$

`\GTSnegDirAdjDown` or $A.\GTSnegDirAdjDown/A.$

`\GTSnDAD` or $A.\GTSnDAD/A.$

will yield $A.\overleftarrow{A}.$

$u \ \GTSnegatedDirectedAdjacencyDown/ \ v$

or $u \ \GTSnegDirAdjDown/ \ v$

or $u \ \GTSnDAD/ \ v$

will yield $u \overleftarrow{} v.$

$\$u \ \GTSnegatedDirectedAdjacencyDown/ \ v\$$

or $\$u \ \GTSnegDirAdjDown/ \ v\$$

or $\$u \ \GTSnDAD/ \ v\$$

will yield $u\overleftarrow{}v.$

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol with relation
`AdjacencyDownRelation` spacing in math mode.

`\GTSnegDirAdjDownRel` $\$u\GTSnegatedDirectedAdjacencyDownRelation/v\$$

`\GTSnDADR` or $\$u\GTSnegDirAdjDownRel/v\$$

or $\$u\GTSnDADR/v\$$

will yield $u \overleftarrow{} v.$

\GTS@i This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and **\GTS@j**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

\{<mainColor> If non-empty, this color will be used as main color.
\{<secondColor> If non-empty, this color will be used as second color.
\{<drawSlash> Non-empty implies that a negation will be drawn.
\{<drawUpArrow> Non-empty implies that an up-arrow will be drawn.
\{<drawDownArrow> Non-empty implies that a down-arrow will be drawn.

\GTS@j This macro builds on top of **\GTS@i** to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

\{<mainColor> If non-empty, this color will be used as main color.
\{<secondColor> If non-empty, this color will be used as second color.
\{<drawSlash> Non-empty implies that a negation will be drawn.
\{<drawUpArrow> Non-empty implies that an up-arrow will be drawn.
\{<drawDownArrow> Non-empty implies that a down-arrow will be drawn.
\{<relation> Non-empty implies that relation spacing will be added.

\GTSxi Versatile macros for incidence that can be given options.

\GTSxinc Argument:

\GTSxincidence [**\{<optionsList>**] Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

\GTSincidence This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

\GTSi **A.\GTSincidence/A.**
or **A.\GTSinc/A.**
or **A.\GTSi/A.**
will yield $A \downarrow A$.
v \GTSincidence/ e
or **v \GTSinc/ e**
or **v \GTSi/ e**
will yield $v \downarrow e$.
\\$v \GTSincidence/ e\\$
or **\\$v \GTSinc/ e\\$**
or **\\$v \GTSi/ e\\$**
will yield $v \downarrow e$.

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.

\GTSincRel **\\$v\GTSincidenceRelation/e\\$**
\GTSiR or **\\$v\GTSincRel/e\\$**
or **\\$v\GTSiR/e\\$**
will yield $v \downarrow e$.

\GTSnegatedIncidence This macro will draw a negated incidence symbol.

\GTSnegInc **A.\GTSnegatedIncidence/A.**
\GTSnI or **A.\GTSnegInc/A.**
or **A.\GTSnI/A.**
will yield $A \not\downarrow A$.
v \GTSnegatedIncidence/ e
or **v \GTSnegInc/ e**
or **v \GTSnI/ e**
will yield $v \not\downarrow e$.

$\$v \backslash \text{GTSnegatedIncidence/ } e\$$
or $\$v \backslash \text{GTSnegInc/ } e\$$
or $\$v \backslash \text{GTSnI/ } e\$$
will yield $v \not\vdash e$.

$\backslash \text{GTSnegatedIncidence-}$ This macro will draw a negated incidence symbol with relation spacing in math mode.

Relation $\$v \backslash \text{GTSnegatedIncidenceRelation/ } e\$$
 $\backslash \text{GTSnegIncRel}$ or $\$v \backslash \text{GTSnegIncRel/ } e\$$
 $\backslash \text{GTSnIR}$ or $\$v \backslash \text{GTSnIR/ } e\$$
will yield $v \not\vdash e$.

$\backslash \text{GTSoutIncidence}$ This macro will draw a directed incidence upward (out(ward) incidence) symbol. The

$\backslash \text{GTSoutInc}$ operand on the left of the symbol should be considered the vertex to which the arc/the
 $\backslash \text{GTSOI}$ operand on the right of the symbol is incident.

$A. \backslash \text{GTSoutIncidence/ } A.$
or $A. \backslash \text{GTSoutInc/ } A.$
or $A. \backslash \text{GTSOI/ } A.$
will yield $A. \vdash A.$
 $v \backslash \text{GTSoutIncidence/ } e$
or $v \backslash \text{GTSoutInc/ } e$
or $v \backslash \text{GTSOI/ } e$
will yield $v \vdash e$.
 $\$v \backslash \text{GTSoutIncidence/ } e\$$
or $\$v \backslash \text{GTSoutInc/ } e\$$
or $\$v \backslash \text{GTSOI/ } e\$$
will yield $v \vdash e$.

$\backslash \text{GTSoutIncidenceRelation}$ This macro will draw a directed incidence upward (out(ward) incidence) symbol with

$\backslash \text{GTSoutIncRel}$ relation spacing in math mode.
 $\backslash \text{GTSOIR}$ $\$v \backslash \text{GTSoutIncidenceRelation/ } e\$$
or $\$v \backslash \text{GTSoutIncRel/ } e\$$
or $\$v \backslash \text{GTSOIR/ } e\$$
will yield $v \vdash e$.

$\backslash \text{GTSnegatedOutIncidence}$ This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

$\backslash \text{GTSnegOutInc}$ $A. \backslash \text{GTSnegatedOutIncidence/ } A.$
 $\backslash \text{GTSnOI}$ or $A. \backslash \text{GTSnegOutInc/ } A.$
or $A. \backslash \text{GTSnOI/ } A.$
will yield $A. \not\vdash A.$
 $v \backslash \text{GTSnegatedOutIncidence/ } e$
or $v \backslash \text{GTSnegOutInc/ } e$
or $v \backslash \text{GTSnOI/ } e$
will yield $v \not\vdash e$.
 $\$v \backslash \text{GTSnegatedOutIncidence/ } e\$$
or $\$v \backslash \text{GTSnegOutInc/ } e\$$
or $\$v \backslash \text{GTSnOI/ } e\$$
will yield $v \not\vdash e$.

$\backslash \text{GTSnegatedOutIncidence-}$ This macro will draw a negated directed incidence upward (out(ward) incidence)

Relation symbol with relation spacing in math mode.
 $\backslash \text{GTSnegOutIncRel}$ $\$v \backslash \text{GTSnegatedOutIncidenceRelation/ } e\$$
 $\backslash \text{GTSnOIR}$ or $\$v \backslash \text{GTSnegOutIncRel/ } e\$$
or $\$v \backslash \text{GTSnOIR/ } e\$$
will yield $v \not\vdash e$.

$\backslash \text{GTSinIncidence}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol. The

$\backslash \text{GTSinInc}$ operand on the left of the symbol should be considered the vertex to which the arc/the
 $\backslash \text{GTSiI}$ operand on the right of the symbol is incident.

$A. \backslash \text{GTSinIncidence/ } A.$
or $A. \backslash \text{GTSinInc/ } A.$
or $A. \backslash \text{GTSiI/ } A.$
will yield $A. \dashv A.$
 $v \backslash \text{GTSinIncidence/ } e$
or $v \backslash \text{GTSinInc/ } e$

or `v \GTSiI/ e`
will yield $v \downarrow e$.
`$v \GTSinIncidence/ e$`
or `$v \GTSinInc/ e$`
or `$v \GTSiI/ e$`
will yield $v \downarrow e$.

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSinIncRel` `$v\GTSinIncidenceRelation/e$`
or `$v\GTSinIncRel/e$`
or `$v\GTSiIR/e$`
will yield $v \downarrow e$.

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.
`\GTSnegInInc`

`\GTSnII` `A.\GTSnegatedInIncidence/A.`
or `A.\GTSnegInInc/A.`
or `A.\GTSnII/A.`
will yield $A \downarrow A$.
`v \GTSnegatedInIncidence/ e`
or `v \GTSnegInInc/ e`
or `v \GTSnII/ e`
will yield $v \downarrow e$.
`$v \GTSnegatedInIncidence/ e$`
or `$v \GTSnegInInc/ e$`
or `$v \GTSnII/ e$`
will yield $v \downarrow e$.

`\GTSnegatedInIncidence-` This macro will draw a negated directed incidence downward (in(ward) incidence) Relation symbol with relation spacing in math mode.

`\GTSnegInIncRel` `$v\GTSnegatedInIncidenceRelation/e$`
`\GTSnIIR` or `$v\GTSnegInIncRel/e$`
or `$v\GTSnIIR/e$`
will yield $v \downarrow e$.

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

`\GTSadjMono` `A.\GTSadjacencyMonochrome/A.`
`\GTSadjMon` or `A.\GTSadjMono/A.`
`\GTSaM` or `A.\GTSadjMon/A.`
or `A.\GTSaM/A.`
will yield $A \circ A$.
`u \GTSadjacencyMonochrome/ v`
or `u \GTSadjMono/ v`
or `u \GTSadjMon/ v`
or `u \GTSaM/ v`
will yield $u \circ v$.
`$u \GTSadjacencyMonochrome/ v$`
or `$u \GTSadjMono/ v$`
or `$u \GTSadjMon/ v$`
or `$u \GTSaM/ v$`
will yield $u \circ v$.

`\GTSadjacencyMonochrome-` This macro will draw a monochrome adjacency symbol with relation spacing in math Relation mode.

`\GTSadjMonoRel` `$u\GTSadjacencyMonochromeRelation/v$`
`\GTSadjMonRel` or `$u\GTSadjMonoRel/v$`
`\GTSaMR` or `$u\GTSadjMonRel/v$`
or `$u\GTSaMR/v$`
will yield $u \circ v$.

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

`\GTSunadjMono` `A.\GTSunadjacencyMonochrome/A.`
`\GTSunadjMon` or `A.\GTSunadjMono/A.`
`\GTSuM`

or A.\GTSunadjMon/A.
 or A.\GTSuM/A.
 will yield $A \circ A$.
 u \GTSunadjacencyMonochrome/ v
 or u \GTSunadjMono/ v
 or u \GTSunadjMon/ v
 or u \GTSuM/ v
 will yield $u \circ v$.
 \$u \GTSunadjacencyMonochrome/ v\$
 or \$u \GTSunadjMono/ v\$
 or \$u \GTSunadjMon/ v\$
 or \$u \GTSuM/ v\$
 will yield $u \circ v$.

\GTSunadjacencyMonochrome- This macro will draw a monochrome unadjacency symbol with relation spacing in
 Relation math mode.

\GTSunadjMonoRel \$u\GTSunadjacencyMonochromeRelation/v\$
 \GTSunadjMonRel or \$u\GTSunadjMonoRel/v\$
 \GTSuMR or \$u\GTSunadjMonRel/v\$
 or \$u\GTSuMR/v\$
 will yield $u \circ v$.

\GTSdirectedAdjacencyUp- This macro will draw a monochrome directed adjacency upward symbol. The operand
 Monochrome on the left of the symbol should be considered the source of the arc; the operand on the
 \GTSdirAdjUpMono right of the symbol should be considered the target of the arc.

\GTSdirAdjUpMon A.\GTSdirectedAdjacencyUpMonochrome/A.
 \GTSdAUM or A.\GTSdirAdjUpMono/A.
 or A.\GTSdirAdjUpMon/A.
 or A.\GTSdAUM/A.
 will yield $A \circ A$.
 u \GTSdirectedAdjacencyUpMonochrome/ v
 or u \GTSdirAdjUpMono/ v
 or u \GTSdirAdjUpMon/ v
 or u \GTSdAUM/ v
 will yield $u \circ v$.
 \$u \GTSdirectedAdjacencyUpMonochrome/ v\$
 or \$u \GTSdirAdjUpMono/ v\$
 or \$u \GTSdirAdjUpMon/ v\$
 or \$u \GTSdAUM/ v\$
 will yield $u \circ v$.

\GTSdirectedAdjacencyUp- This macro will draw a monochrome directed adjacency upward symbol with relation
 MonochromeRelation spacing in math mode.

\GTSdirAdjUpMonoRel \$u\GTSdirectedAdjacencyUpMonochromeRelation/v\$
 \GTSdirAdjUpMonRel or \$u\GTSdirAdjUpMonoRel/v\$
 \GTSdAUMR or \$u\GTSdirAdjUpMonRel/v\$
 or \$u\GTSdAUMR/v\$
 will yield $u \circ v$.

\GTSnegatedDirected- This macro will draw a monochrome negated directed adjacency upward symbol.

AdjacencyUpMonochrome A.\GTSnegatedDirectedAdjacencyUpMonochrome/A.
 \GTSnegDirAdjUpMono or A.\GTSnegDirAdjUpMono/A.
 \GTSnegDirAdjUpMon or A.\GTSnegDirAdjUpMon/A.
 \GTSnDAUM or A.\GTSnDAUM/A.
 will yield $A \circ A$.
 u \GTSnegatedDirectedAdjacencyUpMonochrome/ v
 or u \GTSnegDirAdjUpMono/ v
 or u \GTSnegDirAdjUpMon/ v
 or u \GTSnDAUM/ v
 will yield $u \circ v$.
 \$u \GTSnegatedDirectedAdjacencyUpMonochrome/ v\$
 or \$u \GTSnegDirAdjUpMono/ v\$
 or \$u \GTSnegDirAdjUpMon/ v\$

or $\$u \backslash \text{GTSnDAUM}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash \text{GTSnegatedDirectedAdjacencyUpMonochrome}$ This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

Relation $\$u \backslash \text{GTSnegatedDirectedAdjacencyUpMonochromeRelation}/v\$$
 $\backslash \text{GTSnegDirAdjUpMonoRel}$ or $\$u \backslash \text{GTSnegDirAdjUpMonoRel}/v\$$
 $\backslash \text{GTSnegDirAdjUpMonRel}$ or $\$u \backslash \text{GTSnegDirAdjUpMonRel}/v\$$
 $\backslash \text{GTSnDAUMR}$ or $\$u \backslash \text{GTSnDAUMR}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash \text{GTSdirectedAdjacencyDownMonochrome}$ This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$\backslash \text{GTSdirAdjDownMono}$ $A \backslash \text{GTSdirectedAdjacencyDownMonochrome}/A$.
 $\backslash \text{GTSdADM}$ or $A \backslash \text{GTSdirAdjDownMono}/A$.
or $A \backslash \text{GTSdirAdjDownMon}/A$.
or $A \backslash \text{GTSdADM}/A$.
will yield $A \overset{\circ}{\circ} A$.
 $u \backslash \text{GTSdirectedAdjacencyDownMonochrome}/v$
or $u \backslash \text{GTSdirAdjDownMono}/v$
or $u \backslash \text{GTSdirAdjDownMon}/v$
or $u \backslash \text{GTSdADM}/v$
will yield $u \overset{\circ}{\circ} v$.
 $\$u \backslash \text{GTSdirectedAdjacencyDownMonochrome}/v\$$
or $\$u \backslash \text{GTSdirAdjDownMono}/v\$$
or $\$u \backslash \text{GTSdirAdjDownMon}/v\$$
or $\$u \backslash \text{GTSdADM}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash \text{GTSdirectedAdjacencyDownMonochromeRelation}$ This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

$\backslash \text{GTSdirAdjDownMonoRel}$ $\$u \backslash \text{GTSdirectedAdjacencyDownMonochromeRelation}/v\$$
 $\backslash \text{GTSdirAdjDownMonRel}$ or $\$u \backslash \text{GTSdirAdjDownMonoRel}/v\$$
 $\backslash \text{GTSdADMR}$ or $\$u \backslash \text{GTSdirAdjDownMonRel}/v\$$
or $\$u \backslash \text{GTSdADMR}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash \text{GTSnegatedDirectedAdjacencyDownMonochrome}$ This macro will draw a monochrome negated directed adjacency downward symbol. $A \backslash \text{GTSnegatedDirectedAdjacencyDownMonochrome}/A$.

$\backslash \text{GTSnegDirAdjDownMono}$ or $A \backslash \text{GTSnegDirAdjDownMono}/A$.
 $\backslash \text{GTSnegDirAdjDownMon}$ or $A \backslash \text{GTSnegDirAdjDownMon}/A$.
 $\backslash \text{GTSnDADM}$ or $A \backslash \text{GTSnDADM}/A$.
will yield $A \overset{\circ}{\circ} A$.
 $u \backslash \text{GTSnegatedDirectedAdjacencyDownMonochrome}/v$
or $u \backslash \text{GTSnegDirAdjDownMono}/v$
or $u \backslash \text{GTSnegDirAdjDownMon}/v$
or $u \backslash \text{GTSnDADM}/v$
will yield $u \overset{\circ}{\circ} v$.
 $\$u \backslash \text{GTSnegatedDirectedAdjacencyDownMonochrome}/v\$$
or $\$u \backslash \text{GTSnegDirAdjDownMono}/v\$$
or $\$u \backslash \text{GTSnegDirAdjDownMon}/v\$$
or $\$u \backslash \text{GTSnDADM}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash \text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}$ This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

Relation $\$u \backslash \text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}/v\$$
 $\backslash \text{GTSnegDirAdjDownMonoRel}$ or $\$u \backslash \text{GTSnegDirAdjDownMonoRel}/v\$$
 $\backslash \text{GTSnegDirAdjDownMonRel}$ or $\$u \backslash \text{GTSnegDirAdjDownMonRel}/v\$$
 $\backslash \text{GTSnDADMR}$ or $\$u \backslash \text{GTSnDADMR}/v\$$
will yield $u \overset{\circ}{\circ} v$.

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

`\GTSiM` $A.\GTSincidenceMonochrome/A.$
or $A.\GTSincMono/A.$
or $A.\GTSincMon/A.$
or $A.\GTSiM/A.$
will yield $A.\downarrow A.$
 $v \GTSincidenceMonochrome/ e$
or $v \GTSincMono/ e$
or $v \GTSincMon/ e$
or $v \GTSiM/ e$
will yield $v \downarrow e.$
 $\$v \GTSincidenceMonochrome/ e\$$
or $\$v \GTSincMono/ e\$$
or $\$v \GTSincMon/ e\$$
or $\$v \GTSiM/ e\$$
will yield $v \downarrow e.$

`\GTSincidenceMonochrome-` This macro will draw a monochrome incidence symbol with relation spacing in math
Relation mode.

`\GTSincMonoRel` $\$v\GTSincidenceMonochromeRelation/e\$$
`\GTSincMonRel` or $\$v\GTSincMonoRel/e\$$
`\GTSiMR` or $\$v\GTSincMonRel/e\$$
or $\$v\GTSiMR/e\$$
will yield $v \downarrow e.$

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

`\GTSnegIncMono` $A.\GTSnegatedIncidenceMonochrome/A.$
or $A.\GTSnegIncMono/A.$
`\GTSnegIncMon` or $A.\GTSnegIncMon/A.$
`\GTSnIM` or $A.\GTSnIM/A.$
will yield $A.\downarrow A.$
 $v \GTSnegatedIncidenceMonochrome/ e$
or $v \GTSnegIncMono/ e$
or $v \GTSnegIncMon/ e$
or $v \GTSnIM/ e$
will yield $v \downarrow e.$
 $\$v \GTSnegatedIncidenceMonochrome/ e\$$
or $\$v \GTSnegIncMono/ e\$$
or $\$v \GTSnegIncMon/ e\$$
or $\$v \GTSnIM/ e\$$
will yield $v \downarrow e.$

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing
MonochromeRelation in math mode.

`\GTSnegIncMonoRel` $\$v\GTSnegatedIncidenceMonochromeRelation/e\$$
`\GTSnegIncMonRel` or $\$v\GTSnegIncMonoRel/e\$$
`\GTSnIMR` or $\$v\GTSnegIncMonRel/e\$$
or $\$v\GTSnIMR/e\$$
will yield $v \downarrow e.$

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`\GTSoIM` $A.\GTSoutIncidenceMonochrome/A.$
or $A.\GTSoutIncMono/A.$
or $A.\GTSoutIncMon/A.$
or $A.\GTSoIM/A.$
will yield $A.\uparrow A.$
 $v \GTSoutIncidenceMonochrome/ e$
or $v \GTSoutIncMono/ e$
or $v \GTSoutIncMon/ e$

or $v \setminus \text{GTSoutIM} / e$
will yield $v \uparrow e$.
 $\$v \setminus \text{GTSoutIncidenceMonochrome} / e \$$
or $\$v \setminus \text{GTSoutIncMono} / e \$$
or $\$v \setminus \text{GTSoutIncMon} / e \$$
or $\$v \setminus \text{GTSoutIM} / e \$$
will yield $v \uparrow e$.

$\setminus \text{GTSoutIncidenceMonochrome-}$ This macro will draw a monochrome directed incidence upward (out(ward) incidence)
Relation symbol with relation spacing in math mode.
 $\setminus \text{GTSoutIncMonoRel} \ \$v \setminus \text{GTSoutIncidenceMonochromeRelation} / e \$$
 $\setminus \text{GTSoutIncMonRel} \ \text{or } \$v \setminus \text{GTSoutIncMonoRel} / e \$$
 $\setminus \text{GTSoutIMR} \ \text{or } \$v \setminus \text{GTSoutIncMonRel} / e \$$
or $\$v \setminus \text{GTSoutIMR} / e \$$
will yield $v \uparrow e$.

$\setminus \text{GTSnegatedOutIncidence-}$ This macro will draw a monochrome negated directed incidence upward (out(ward)
Monochrome incidence) symbol.
 $\setminus \text{GTSnegOutIncMono} \ A. \setminus \text{GTSnegatedOutIncidenceMonochrome} / A.$
 $\setminus \text{GTSnegOutIncMon} \ \text{or } A. \setminus \text{GTSnegOutIncMono} / A.$
 $\setminus \text{GTSnOIM} \ \text{or } A. \setminus \text{GTSnegOutIncMon} / A.$
or $A. \setminus \text{GTSnOIM} / A.$
will yield $A. \uparrow A.$
 $v \setminus \text{GTSnegatedOutIncidenceMonochrome} / e$
or $v \setminus \text{GTSnegOutIncMono} / e$
or $v \setminus \text{GTSnegOutIncMon} / e$
or $v \setminus \text{GTSnOIM} / e$
will yield $v \uparrow e$.
 $\$v \setminus \text{GTSnegatedOutIncidenceMonochrome} / e \$$
or $\$v \setminus \text{GTSnegOutIncMono} / e \$$
or $\$v \setminus \text{GTSnegOutIncMon} / e \$$
or $\$v \setminus \text{GTSnOIM} / e \$$
will yield $v \uparrow e$.

$\setminus \text{GTSnegatedOutIncidence-}$ This macro will draw a monochrome negated directed incidence upward (out(ward)
MonochromeRelation incidence) symbol with relation spacing in math mode.
 $\setminus \text{GTSnegOutIncMonoRel} \ \$v \setminus \text{GTSnegatedOutIncidenceMonochromeRelation} / e \$$
 $\setminus \text{GTSnegOutIncMonRel} \ \text{or } \$v \setminus \text{GTSnegOutIncMonoRel} / e \$$
 $\setminus \text{GTSnOIMR} \ \text{or } \$v \setminus \text{GTSnegOutIncMonRel} / e \$$
or $\$v \setminus \text{GTSnOIMR} / e \$$
will yield $v \uparrow e$.

$\setminus \text{GTSinIncidenceMonochrome}$ This macro will draw a monochrome directed incidence downward (in(ward) incidence)
 $\setminus \text{GTSinIncMono}$ symbol. The operand on the left of the symbol should be considered the vertex to which
 $\setminus \text{GTSinIncMon}$ the arc/the operand on the right of the symbol is incident.
 $\setminus \text{GTSiIM} \ A. \setminus \text{GTSinIncidenceMonochrome} / A.$
or $A. \setminus \text{GTSinIncMono} / A.$
or $A. \setminus \text{GTSinIncMon} / A.$
or $A. \setminus \text{GTSiIM} / A.$
will yield $A. \downarrow A.$
 $v \setminus \text{GTSinIncidenceMonochrome} / e$
or $v \setminus \text{GTSinIncMono} / e$
or $v \setminus \text{GTSinIncMon} / e$
or $v \setminus \text{GTSiIM} / e$
will yield $v \downarrow e$.
 $\$v \setminus \text{GTSinIncidenceMonochrome} / e \$$
or $\$v \setminus \text{GTSinIncMono} / e \$$
or $\$v \setminus \text{GTSinIncMon} / e \$$
or $\$v \setminus \text{GTSiIM} / e \$$
will yield $v \downarrow e$.

$\setminus \text{GTSinIncidenceMonochrome-}$ This macro will draw a monochrome directed incidence downward (in(ward) incidence)
Relation symbol with relation spacing in math mode.
 $\setminus \text{GTSinIncMonoRel} \ \$v \setminus \text{GTSinIncidenceMonochromeRelation} / e \$$
 $\setminus \text{GTSinIncMonRel}$
 $\setminus \text{GTSiIMR}$

or $\$v\backslash\text{GTSinIncMonoRel}/e\$$
or $\$v\backslash\text{GTSinIncMonRel}/e\$$
or $\$v\backslash\text{GTSiIMR}/e\$$
will yield $v \not\vdash e$.

$\backslash\text{GTSnegatedInIncidence-}$ This macro will draw a monochrome negated directed incidence downward (in(ward)
Monochrome incidence) symbol.

$\backslash\text{GTSnegInIncMono}$ $A.\backslash\text{GTSnegatedInIncidenceMonochrome}/A.$

$\backslash\text{GTSnegInIncMon}$ or $A.\backslash\text{GTSnegInIncMono}/A.$

$\backslash\text{GTSnIIM}$ or $A.\backslash\text{GTSnegInIncMon}/A.$

or $A.\backslash\text{GTSnIIM}/A.$

will yield $A.\not\vdash A.$

$v \backslash\text{GTSnegatedInIncidenceMonochrome}/ e$

or $v \backslash\text{GTSnegInIncMono}/ e$

or $v \backslash\text{GTSnegInIncMon}/ e$

or $v \backslash\text{GTSnIIM}/ e$

will yield $v \not\vdash e$.

$\$v \backslash\text{GTSnegatedInIncidenceMonochrome}/ e\$$

or $\$v \backslash\text{GTSnegInIncMono}/ e\$$

or $\$v \backslash\text{GTSnegInIncMon}/ e\$$

or $\$v \backslash\text{GTSnIIM}/ e\$$

will yield $v \not\vdash e$.

$\backslash\text{GTSnegatedInIncidence-}$ This macro will draw a monochrome negated directed incidence downward (in(ward)
MonochromeRelation incidence) symbol with relation spacing in math mode.

$\backslash\text{GTSnegInIncMonoRel}$ $\$v\backslash\text{GTSnegatedInIncidenceMonochromeRelation}/e\$$

$\backslash\text{GTSnegInIncMonRel}$ or $\$v\backslash\text{GTSnegInIncMonoRel}/e\$$

$\backslash\text{GTSnIIMR}$ or $\$v\backslash\text{GTSnegInIncMonRel}/e\$$

or $\$v\backslash\text{GTSnIIMR}/e\$$

will yield $v \not\vdash e$.

$\backslash\text{GTSmetaNot}$ This macro will draw a meta-not or relation-not symbol.

$\backslash\text{GTSmN}$ $A.\backslash\text{GTSmetaNot}/A.$

or $A.\backslash\text{GTSmN}/A.$

will yield $A.\not\models A.$

$u \backslash\text{GTSmetaNot}/\backslash\text{GTSdAU}/ v$

or $u \backslash\text{GTSmN}/\backslash\text{GTSdAU}/ v$

will yield $u \not\models v$.

$\$u \backslash\text{GTSmetaNot}/ \backslash\text{GTSdAU}/ v\$$

or $\$u \backslash\text{GTSmN}/ \backslash\text{GTSdAU}/ v\$$

will yield $u \not\models v$.

$\backslash\text{GTSmetaAnd}$ This macro will draw a meta-and or relations-and symbol. This is your duty to check
 $\backslash\text{GTSmA}$ that all relations have same arity. It should be easy as long as you remain in the realm of
graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup
has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge
symbols have taken various meanings, whilst we want an unambiguous symbol.
Look at the discussion after all logical symbols for less trivial uses and recommendations.

$A.\backslash\text{GTSmetaAnd}/A.$

or $A.\backslash\text{GTSmA}/A.$

will yield $A.\not\wedge A.$

$u \backslash\text{GTSdAD}/\backslash\text{GTSmetaAnd}/\backslash\text{GTSdAU}/ v$

or $u \backslash\text{GTSdAD}/\backslash\text{GTSmA}/\backslash\text{GTSdAU}/ v$

will yield $u \not\wedge v$.

$\$u \backslash\text{GTSdAD}/ \backslash\text{GTSmetaAnd}/ \backslash\text{GTSdAU}/ v\$$

or $\$u \backslash\text{GTSdAD}/ \backslash\text{GTSmA}/ \backslash\text{GTSdAU}/ v\$$

will yield $u \not\wedge v$.

$\backslash\text{GTSmetaAndRelation}$ This macro will draw a meta-and symbol with relation spacing in math mode.

$\backslash\text{GTSmetaAndRel}$ $\$u \backslash\text{GTSdAD}/\backslash\text{GTSmetaAndRelation}/\backslash\text{GTSdAU}/ v\$$

$\backslash\text{GTSmAR}$ or $\$u \backslash\text{GTSdAD}/\backslash\text{GTSmetaAndRel}/\backslash\text{GTSdAU}/ v\$$

or $\$u \backslash\text{GTSdAD}/\backslash\text{GTSmAR}/\backslash\text{GTSdAU}/ v\$$

will yield $u \not\wedge v$.

`\GTSmetaOr` This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol. Look at the discussion after all logical symbols for less trivial uses and recommendations.

`A.\GTSmetaOr/A.`
or `A.\GTSmO/A.`
will yield $A.\bigvee A.$
`u \GTSdAD/\GTSmetaOr/\GTSdAU/ v`
or `u \GTSdAD/\GTSmO/\GTSdAU/ v`
will yield $u \bigvee v.$
`$u \GTSdAD/ \GTSmetaOr/ \GTSdAU/ v$`
or `$u \GTSdAD/ \GTSmO/ \GTSdAU/ v$`
will yield $u \bigvee v.$

`\GTSmetaOrRelation` This macro will draw a meta-or symbol with relation spacing in math mode.

`\GTSmetaOrRel $u \GTSdAD/\GTSmetaOrRelation/\GTSdAU/ v$`
`\GTSmOR` or `$u \GTSdAD/\GTSmetaOrRel/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmOR/\GTSdAU/ v$`
will yield $u \bigvee v.$

2.3 Good practices

As is customary in algebra and logic, the conjunction/multiplication can be replaced by concatenation, and has higher priority than disjunction. So the following formulas are equivalent; we recommend using the shortest one:

Tournament adjacency:

$\forall u, v, u \bigvee v$

is preferred over

$\forall u, v, u \bigvee \bigwedge v,$

or $\forall u, v, u (\bigvee \bigwedge) \bigvee (\bigwedge \bigvee) v.$

Note the use of macros without Relation suffix above, and adding a single `\mathrel{}` around the composition of all relations:

`\mathrel{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/}`.

This is the spacing practice that we recommend, although for very long compositions, it may be quite unreadable, and then spacing everything with Relation could/should be preferred.

We also recommend using brackets when the composition reflects all of the adjacencies between two vertices:

Tournament adjacency:

$\forall u, v, u [\bigvee \bigwedge] v.$

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u [\bigvee \bigwedge] v.$

Directed graph:

$\forall u, v, u [\bigvee \bigwedge \bigvee \bigwedge] v.$

Oriented graph:

$\forall u, v, u [\bigvee \bigwedge \bigvee \bigwedge] v.$

Finally, we recommend prefix notation without “big” symbol for a sequence of consecutive ors or ands (with parentheses if needed):

$\exists u, v, u [\bigvee \bigwedge] v.$

$\exists v, e, v [\bigvee \bigwedge] e.$

3 Implementation

```
1 \<package>
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
```

```

8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}
14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17
\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{\commandname} The name of the command to create.
{\commandcode} The code of the command to create.
18 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
19 \@ifdefinable{#1}{\def#1/{#2}}
20 }}

\GTS@newprotectedcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{\commandname} The name of the command to create.
{\commandcode} The code of the command to create.
21 \@ifdefinable{\GTS@newprotectedcommand}{\def\GTS@newprotectedcommand#1#2{%
22 \@ifdefinable{#1}{\protected\def#1/{#2}}
23 }}

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value
“black”.
Argument:
{\color} The color that must be used by GraphTheorySymbol afterward.
24 \newcommand{\GTSsetMainColor}[1]{%
25 \edef\GTS@mainColorTemp{#1}%
26 \edef\GTS@mainColorBackup{\GTS@mainColor}%
27 \edef\GTS@mainColor{\GTS@mainColorTemp}%
28 }

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial
value “gray”.
Argument:
{\color} The color that must be used by GraphTheorySymbol afterward.
29 \newcommand{\GTSsetSecondColor}[1]{%
30 \edef\GTS@secondColorTemp{#1}%
31 \edef\GTS@secondColorBackup{\GTS@secondColor}%
32 \edef\GTS@secondColor{\GTS@secondColorTemp}%
33 }

\GTS@a This macro will draw any adjacency symbol given the correct parameters. Its use is purely
technical. See the macros derived from it and \GTS@b: \GTSxadjacency, \GTSxadj, and
\GTSxa.
Arguments:
{\mainColor} If non-empty, this color will be used as main color.
{\secondColor} If non-empty, this color will be used as second color.
{\drawSlash} Non-empty implies that a negation will be drawn.
{\drawUpArrow} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow} Non-empty implies that a down-arrow will be drawn.
34 \newcommand{\GTS@a}[5]{%
35 \if#1\empty\else%
36 \GTSsetMainColor{#1}%
37 \fi%
38 \if#2\empty\else%
39 \GTSsetSecondColor{#2}%

```



```

40 \fi%
41 \text{\ensuremath{%
42 \raisebox{-0.32ex}{\scalebox{1.5}[1.5]{%
43 %
44 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
45 $\mathcolor{\GTS@secondColor}{\circ}$%
46 }}%
47 \hspace{-0.39ex}%
48 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
49 $\mathcolor{\GTS@secondColor}{\bullet}$%
50 }}%
51 \hspace{-0.12ex}%
52 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
53 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
54 }}%
55 \hspace{-0.40ex}%
56 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
57 $\mathcolor{\GTS@secondColor}{\circ}$%
58 }}%
59 \hspace{-0.39ex}%
60 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
61 $\mathcolor{\GTS@secondColor}{\bullet}$%
62 }}%
63 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
64 \if#4\empty\else%
65 \hspace{-0.65ex}%
66 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
67 \textcolor{\GTS@mainColor}{\textasciicircum}%
68 }}%
69 \fi%
if GTS@drawDownArrow empty else
70 \if#5\empty\else%
71 \hspace{-0.65ex}%
72 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
73 \textcolor{\GTS@mainColor}{\textasciicircum}%
74 }}%
75 \fi%
if GTS@drawSlash empty else
76 \if#3\empty\else%
77 \hspace{-0.65ex}%
78 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
79 \textcolor{black}{/}%
80 }}%
81 \fi%
82 %
83 }}%
84 }}%
85 \if#1\empty\else%
86 \GTSsetMainColor{\GTS@mainColorBackup}%
87 \fi%
88 \if#2\empty\else%
89 \GTSsetSecondColor{\GTS@secondColorBackup}%
90 \fi%
91 }

```

\GTS@b This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

{*mainColor*} If non-empty, this color will be used as main color.

{*secondColor*} If non-empty, this color will be used as second color.

{*drawSlash*} Non-empty implies that a negation will be drawn.

{*drawUpArrow*} Non-empty implies that an up-arrow will be drawn.

$\{\langle drawDownArrow \rangle\}$ Non-empty implies that a down-arrow will be drawn.
 $\{\langle relation \rangle\}$ Non-empty implies that relation spacing will be added.

```

92 \newcommand{\GTS@b}[6]{%
93 \if#61%
94 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
95 \else%
96 \GTS@a{#1}{#2}{#3}{#4}{#5}%
97 \fi%
98 }

```

$\backslash\text{GTSxa}$ Versatile macros for adjacency that can be given options.

$\backslash\text{GTSxadj}$ Argument:

$\backslash\text{GTSxadjacency}$ [$\langle optionsList \rangle$] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

```

99 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
101 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
102 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
103 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
105 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
106 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
107 \define@key{GTSxa}{monochrome}[1]%
108 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
109 \define@key{GTSxa}{mono}[1]%
110 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
111 \define@key{GTSxa}{mon}[1]%
112 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
113 \define@key{GTSxa}{m}[1]%
114 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
115 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
116 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
117 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
118 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
119 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
120 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
121 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
122 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
123 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
124 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
125
126 \newcommand{\GTSxa}[1] [] {%
127 \def\GTS@@mainColor{}%
128 \def\GTS@@secondColor{}%
129 \def\GTS@@drawSlash{}%
130 \def\GTS@@drawUpArrow{}%
131 \def\GTS@@drawDownArrow{}%
132 \def\GTS@@relation{}%
133 \setkeys{GTSxa}{#1}%
134 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
135 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
136 }
137
138 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
139
140 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

$\backslash\text{GTSadjacency}$ This macro will draw an adjacency symbol.

$\backslash\text{GTSadj}$ 141 $\backslash\text{GTS@newcommand}\{\backslash\text{GTSadjacency}\}\{\backslash\text{GTS@a}\{\}\{\}\{\}\}$
 $\backslash\text{GTSa}$

```

142 \GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}
143 \GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}

```

`\GTSadjacencyRelation` This macro will draw an adjacency symbol with relation spacing in math mode.

```

\GTSadjRel 144 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
\GTSaR 145 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
146 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

```

`\GTSunadjacency` This macro will draw an unadjacency symbol.

```

\GTSunadj 147 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}{}}
\GTSu 148 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}{}}
149 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}{}}

```

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol with relation spacing in math mode.

```

\GTSunadjRel 150 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
\GTSuR 151 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
152 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

153 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}{}}
154 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}{}}
155 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}{}}

```

`\GTSdirectedAdjacencyUp-Relation` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

\GTSdirAdjUpRel 156 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
\GTSdAUR 157 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
158 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

`\GTSnegatedDirected-AdjacencyUp` This macro will draw a negated directed adjacency upward symbol.

```

159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}{}}
\GTSnegDirAdjUp 160 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}{}}
\GTSnDAU 161 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}{}}

```

`\GTSnegatedDirected-AdjacencyUpRelation` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

\GTSnegDirAdjUpRel 162 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
\GTSnDAUR 163 \mathrel{\GTSnDAU/}%
164 }
165 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
166 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

```

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

167 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}{}{}}
168 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}{}{}}
169 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}{}{}}

```

`\GTSdirectedAdjacencyDown-Relation` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdirAdjDownRel 170 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
\GTSdADR 171 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
172 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

```

`\GTSnegatedDirected-AdjacencyDown` This macro will draw a negated directed adjacency downward symbol.

```

173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{}{1}{}{}}
\GTSnegDirAdjDown 174 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{}{1}{}{}}
\GTSnDAD 175 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{}{1}{}{}}

```

`\GTSnegatedDirectedAdjacencyDownRelation` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

\GTSnegDirAdjDownRel 176 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
\GTSnDADR 177 \mathrel{\GTSnDAD/}%
178 }
179 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
180 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

```

`\GTS@i` This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.
`{\secondColor}` If non-empty, this color will be used as second color.
`{\drawSlash}` Non-empty implies that a negation will be drawn.
`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.
`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

```

181 \newcommand{\GTS@i}[5]{%
182 \if#1\empty\else%
183 \GTSsetMainColor{#1}%
184 \fi%
185 \if#2\empty\else%
186 \GTSsetSecondColor{#2}%
187 \fi%
188 \text{\ensuremath{%
189 \raisebox{-0.22ex}{\scalebox{2.05}[2.05]{%
190 %
191 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
192 $\mathcolor{\GTS@secondColor}{\circ}$%
193 }}%
194 \hspace{-0.39ex}%
195 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
196 $\mathcolor{\GTS@secondColor}{\bullet}$%
197 }}%
198 \hspace{-0.12ex}%
199 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
200 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
201 }}%
202 \hspace{-0.18ex}%
203 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
204 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
205 }}%
206 \hspace{0.2ex}%
207 \if#4\empty\else%
208 \hspace{-0.65ex}%
209 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
210 \textcolor{\GTS@secondColor}{\textasciicircum}%
211 }}%
212 \fi%
213 \if#5\empty\else%
214 \hspace{-0.65ex}%
215 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
216 \textcolor{\GTS@secondColor}{\textasciicircum}%
217 }}%
218 \fi%
219 \if#3\empty\else%
220 \hspace{-0.65ex}%
221 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
222 \textcolor{black}{/}%

```

```

223 }}%
224 \fi%
225 %
226 }}%
227 }}%
228 \if#1\empty\else%
229 \GTSsetMainColor{\GTS@mainColorBackup}%
230 \fi%
231 \if#2\empty\else%
232 \GTSsetSecondColor{\GTS@secondColorBackup}%
233 \fi%
234 }

```

\GTS@j This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

{<mainColor>} If non-empty, this color will be used as main color.
{<secondColor>} If non-empty, this color will be used as second color.
{<drawSlash>} Non-empty implies that a negation will be drawn.
{<drawUpArrow>} Non-empty implies that an up-arrow will be drawn.
{<drawDownArrow>} Non-empty implies that a down-arrow will be drawn.
{<relation>} Non-empty implies that relation spacing will be added.

```

235 \newcommand{\GTS@j}[6]{%
236 \if#61%
237 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
238 \else%
239 \GTS@i{#1}{#2}{#3}{#4}{#5}%
240 \fi%
241 }

```

\GTSxi Versatile macros for incidence that can be given options.

\GTSxinc Argument:

\GTSxincidence [*{<optionsList>}*] Sets the following options if not empty:
 mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
 secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
 monochrome or mono or mon or m (equivalent to mC=black,sC=black),
 negated or not or n,
 in or i,
 out or o,
 relation or rel or r.

```

242 \define@key{GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
243 \define@key{GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
244 \define@key{GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
245 \define@key{GTSxi}{mC}{\def\GTS@@mainColor{#1}}
246 \define@key{GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
247 \define@key{GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
248 \define@key{GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
249 \define@key{GTSxi}{sC}{\def\GTS@@secondColor{#1}}
250 \define@key{GTSxi}{monochrome}[1]%
251 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
252 \define@key{GTSxi}{mono}[1]%
253 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
254 \define@key{GTSxi}{mon}[1]%
255 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
256 \define@key{GTSxi}{m}[1]%
257 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
258 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
259 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
260 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
261 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
262 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
263 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
264 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}

```

```

265 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
266 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
267 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
268
269 \newcommand{\GTSxi}[1][]{\%
270 \def\GTS@@mainColor{}}
271 \def\GTS@@secondColor{}}
272 \def\GTS@@drawSlash{}}
273 \def\GTS@@drawUpArrow{}}
274 \def\GTS@@drawDownArrow{}}
275 \def\GTS@@relation{}}
276 \setkeys{GTSxi}{#1}}
277 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
278 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
279 }
280
281 \newcommand{\GTSxinc}[1][]{\GTSxi[#1]}
282
283 \newcommand{\GTSxincidence}[1][]{\GTSxi[#1]}

```

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

\GTSinc 284 \GTS@newcommand{\GTSincidence}{\GTS@i{}{}{}{}}
\GTSi    285 \GTS@newcommand{\GTSinc}{\GTS@i{}{}{}{}}
          286 \GTS@newcommand{\GTSi}{\GTS@i{}{}{}{}}

```

`\GTSincidenceRelation` This macro will draw an incidence symbol with relation spacing in math mode.

```

\GTSincRel 287 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
\GTSiR      288 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
          289 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

```

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.

```

\GTSnegInc 290 \GTS@newcommand{\GTSnegatedIncidence}{\GTS@i{}{}{1}{}{}}
\GTSnI      291 \GTS@newcommand{\GTSnegInc}{\GTS@i{}{}{1}{}{}}
          292 \GTS@newcommand{\GTSnI}{\GTS@i{}{}{1}{}{}}

```

`\GTSnegatedIncidence-` This macro will draw a negated incidence symbol with relation spacing in math mode.

```

Relation    293 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
\GTSnegIncRel 294 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
\GTSnIR      295 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

```

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

296 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{1}{}{}}
297 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{1}{}{}}
298 \GTS@newcommand{\GTSOI}{\GTS@i{}{}{1}{}{}}

```

`\GTSoutIncidenceRelation` This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

\GTSoutIncRel 299 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSOI/}}
\GTSOIIR      300 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSOI/}}
          301 \GTS@newcommand{\GTSOIIR}{\mathrel{\GTSOI/}}

```

`\GTSnegatedOutIncidence` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```

\GTSnegOutInc 302 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}{}}
\GTSnOI        303 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}{}}
          304 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}{}}

```

`\GTSnegatedOutIncidence-` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

Relation      305 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
\GTSnOIR      306 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
          307 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}

```

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the

`\GTSinInc` operand on the right of the symbol is incident.

```

308 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{1}}
309 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{1}}
310 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{1}}

```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSiIR`

```

311 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
312 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
313 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

`\GTSnegInInc`

`\GTSnII`

```

314 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{}{1}}
315 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{}{1}}
316 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{}{1}}

```

`\GTSnegatedInIncidence-Relation` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSnegInIncRel`

`\GTSnIIR`

```

317 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
318 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
319 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}

```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

`\GTSadjMono`

`\GTSadjMon`

`\GTSaM`

```

320 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}{}{}}
321 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}{}{}}
322 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}{}{}}
323 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}{}{}}

```

`\GTSadjacencyMonochrome-Relation` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

`\GTSadjMonoRel`

`\GTSadjMonRel`

`\GTSaMR`

```

324 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
325 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
326 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
327 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}

```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

`\GTSunadjMono`

`\GTSunadjMon`

`\GTSuM`

```

328 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}{}}
329 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}{}}
330 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}{}}
331 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}{}}

```

`\GTSunadjacencyMonochrome-Relation` This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

`\GTSunadjMonoRel`

`\GTSunadjMonRel`

`\GTSuMR`

```

332 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
333 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
334 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
335 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}

```

`\GTSdirectedAdjacencyUpMonochrome` This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

`\GTSdirAdjUpMono`

`\GTSdirAdjUpMon`

`\GTSdAUM`

```

336 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
337 \GTS@a{black}{black}{}{1}{}%
338 }
339 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{}{1}{}{}}
340 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{}{1}{}{}}
341 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{}{1}{}{}}

```

`\GTSdirectedAdjacencyUpMonochromeRelation` This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

`\GTSdirAdjUpMonoRel`

`\GTSdirAdjUpMonRel`

`\GTSdAUMR`

```

342 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
343 \mathrel{\GTSdAUM/}%
344 }

```

```

345 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
346 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
347 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}

```

`\GTSnegatedDirectedAdjacencyUpMonochrome` This macro will draw a monochrome negated directed adjacency upward symbol.

```

348 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
\GTSnegDirAdjUpMono 349 \GTS@a{black}{black}{1}{1}{}%
\GTSnegDirAdjUpMon 350 }
\GTSnDAUM 351 \GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{}}
352 \GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{}}
353 \GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{}}

```

`\GTSnegatedDirectedAdjacencyUpMonochromeRelation` This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```

Relation 354 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
\GTSnegDirAdjUpMonRel 355 \mathrel{\GTSnDAUM/}%
\GTSnegDirAdjUpMonoRel 356 }
\GTSnDAUMR 357 \GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\GTSnDAUM/}}
358 \GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\GTSnDAUM/}}
359 \GTS@newcommand{\GTSnDAUMR}{\mathrel{\GTSnDAUM/}}

```

`\GTSdirectedAdjacencyDownMonochrome` This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

\GTSdirAdjDownMono 360 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
\GTSdADM 361 \GTS@a{black}{black}{-}{-}{1}%
362 }
363 \GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{-}{-}{1}}
364 \GTS@newcommand{\GTSdirAdjDownMon}{\GTS@a{black}{black}{-}{-}{1}}
365 \GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{-}{-}{1}}

```

`\GTSdirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdirAdjDownMonoRel 366 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
\GTSdirAdjDownMonRel 367 \mathrel{\GTSdADM/}%
\GTSdADMR 368 }
369 \GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\GTSdADM/}}
370 \GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\GTSdADM/}}
371 \GTS@newcommand{\GTSdADMR}{\mathrel{\GTSdADM/}}

```

`\GTSnegatedDirectedAdjacencyDownMonochrome` This macro will draw a monochrome negated directed adjacency downward symbol.

```

372 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
\GTSnegDirAdjDownMono 373 \GTS@a{black}{black}{1}{-}{1}%
\GTSnegDirAdjDownMon 374 }
\GTSnDADM 375 \GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{-}{1}}
376 \GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{-}{1}}
377 \GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{-}{1}}

```

`\GTSnegatedDirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```

Relation 378 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
\GTSnegDirAdjDownMonoRel 379 \mathrel{\GTSnDADM/}%
\GTSnegDirAdjDownMonRel 380 }
\GTSnDADMR 381 \GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\GTSnDADM/}}
382 \GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\GTSnDADM/}}
383 \GTS@newcommand{\GTSnDADMR}{\mathrel{\GTSnDADM/}}

```

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

\GTSincMono 384 \GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{-}{-}{-}}
\GTSincMon 385 \GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{-}{-}{-}}
386 \GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{-}{-}{-}}
387 \GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{-}{-}{-}}

```


`\GTSincidenceMonochrome-` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

Relation 388 `\GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\GTSiM/}}`

`\GTSincMonoRel` 389 `\GTS@newcommand{\GTSincMonoRel}{\mathrel{\GTSiM/}}`

`\GTSincMonRel` 390 `\GTS@newcommand{\GTSincMonRel}{\mathrel{\GTSiM/}}`

`\GTSiMR` 391 `\GTS@newcommand{\GTSiMR}{\mathrel{\GTSiM/}}`

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

Monochrome 392 `\GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%`

`\GTSnegIncMono` 393 `\GTS@i{black}{black}{1}{1}{1}{1}%`

`\GTSnegIncMon` 394 `}`

`\GTSnIM` 395 `\GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{1}{1}{1}}`

396 `\GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{1}{1}{1}}`

397 `\GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{1}{1}{1}}`

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

MonochromeRelation 398 `\GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}`

`\GTSnegIncMonoRel` 399 `\GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}`

`\GTSnegIncMonRel` 400 `\GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}`

`\GTSnIMR` 401 `\GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}`

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the

`\GTSoutIncMono` arc/the operand on the right of the symbol is incident.

`\GTSoutIncMon` 402 `\GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{1}{1}{1}{1}}`

403 `\GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{1}{1}{1}{1}}`

404 `\GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{1}{1}{1}{1}}`

405 `\GTS@newcommand{\GTSoIM}{\GTS@i{black}{black}{1}{1}{1}{1}}`

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

Relation 406 `\GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoIM/}}`

`\GTSoutIncMonoRel` 407 `\GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoIM/}}`

`\GTSoutIncMonRel` 408 `\GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoIM/}}`

`\GTSoIMR` 409 `\GTS@newcommand{\GTSoIMR}{\mathrel{\GTSoIM/}}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

Monochrome 410 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%`

`\GTSnegOutIncMono` 411 `\GTS@i{black}{black}{1}{1}{1}{1}%`

`\GTSnegOutIncMon` 412 `}`

`\GTSnOIM` 413 `\GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}{1}{1}}`

414 `\GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}{1}{1}}`

415 `\GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}{1}{1}}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

MonochromeRelation 416 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%`

`\GTSnegOutIncMonoRel` 417 `\mathrel{\GTSnOIM/}%`

`\GTSnegOutIncMonRel` 418 `}`

`\GTSnOIMR` 419 `\GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}`

420 `\GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}`

421 `\GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}`

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the

`\GTSinIncMono` arc/the operand on the right of the symbol is incident.

`\GTSinIncMon` 422 `\GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{1}{1}{1}{1}}`

`\GTSiIM` 423 `\GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{1}{1}{1}{1}}`

424 `\GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{1}{1}{1}{1}}`

425 `\GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{1}{1}{1}{1}}`

`\GTSinIncidenceMonochromeRelation` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

\GTSinIncMonoRel 426 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}
\GTSinIncMonRel 427 \GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}
\GTSiIMR 428 \GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}
429 \GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}

```

`\GTSnegatedInIncidenceMonochrome` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```

\GTSnegInIncMono 430 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
\GTSnegInIncMon 431 \GTS@i{black}{black}{1}{1}%
\GTSnIIM 432 }
433 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{1}}
434 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{1}}
435 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{1}}

```

`\GTSnegatedInIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

\GTSnegInIncMonoRel 436 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
\GTSnegInIncMonRel 437 \mathrel{\GTSnIIM/}%
\GTSnIIMR 438 }
439 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
440 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
441 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}

```

`\GTSmetaNot` This macro will draw a meta-not or relation-not symbol.

```

\GTSmN 442 \GTS@newprotectedcommand{\GTSmN}{%
443 \text{\ensuremath{%
444 \raisebox{-0.44ex}{\scalebox{1}{1}}{%
445 \raisebox{0ex}{\scalebox{1}{1}}{\$ \lnot \$}}%
446 \hspace{-1.46ex}%
447 \raisebox{1.2ex}{\scalebox{0.7}{0.7}{m}}%
448 \hspace{0.2ex}%
449 }}%
450 }}%
451 }
452 \GTS@newcommand{\GTSmetaNot}{\GTSmN/}

```

`\GTSmetaAnd` This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

```

\GTSmA 453 \GTS@newprotectedcommand{\GTSmA}{%
454 \text{\ensuremath{%
455 \raisebox{-0.44ex}{\scalebox{0.95}{0.95}}{%
456 \raisebox{0ex}{\scalebox{1}{1}}{\$ \wedge \$}}%
457 \hspace{-1.46ex}%
458 \raisebox{1.5ex}{\scalebox{0.7}{0.7}{m}}%
459 }}%
460 }}%
461 }
462 \GTS@newcommand{\GTSmetaAnd}{\GTSmA/}

```

`\GTSmetaAndRelation` This macro will draw a meta-and symbol with relation spacing in math mode.

```

\GTSmetaAndRel 463 \GTS@newcommand{\GTSmetaAndRelation}{\mathrel{\GTSmA/}}
\GTSmAR 464 \GTS@newcommand{\GTSmetaAndRel}{\mathrel{\GTSmA/}}
465 \GTS@newcommand{\GTSmAR}{\mathrel{\GTSmA/}}

```

`\GTSmetaOr` This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures ;). We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublevee or other vee symbols have taken various meanings, whilst we want an unambiguous symbol.

```

\GTSmO

```

```

466 \GTS@newprotectedcommand{\GTSmO}{%
467 \text{\ensuremath{%
468 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
469 \raisebox{0ex}{\scalebox{1}[1]{\$vee$}}}%
470 \hspace{-1.46ex}}%
471 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
472 }}%
473 }}%
474 }
475 \GTS@newcommand{\GTSMetaOr}{\GTSmO/}

```

`\GTSMetaOrRelation` This macro will draw a meta-or symbol with relation spacing in math mode.

```

\GTSMetaOrRel 476 \GTS@newcommand{\GTSMetaOrRelation}{\mathrel{\GTSmO/}}
\GTSMOR 477 \GTS@newcommand{\GTSMetaOrRel}{\mathrel{\GTSmO/}}
478 \GTS@newcommand{\GTSMOR}{\mathrel{\GTSmO/}}

479 \endinput
480 \</package>

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

6 Change History

v0.1.0 – 2026-08-01	same set of symbols	1
General: Initial version	1	
v1.0.0 – 2026-08-26	v1.2.1 – 2026-09-08	
General: First public release	General: smaller tests output	1
v1.1.0 – 2026-08-29	v1.3.0 – 2026-09-08	
General: Added a memo	General: new macros for monochrome symbols	1
v1.1.1 – 2026-08-30	v1.4.0 – 2026-09-12	
General: Small corrections and improvements	General: macros for symbols for logical operations on relations	1
v1.2.0 – 2026-09-06		
General: CTAN URL, new macros for		

Index

G

<code>\GTS@@drawDownArrow</code>	120, 121, 131, 135, 263, 264, 274, 278
<code>\GTS@@drawSlash</code>	115, 116, 117, 129, 134, 258, 259, 260, 272, 277
<code>\GTS@@drawUpArrow</code>	118, 119, 130, 135, 261, 262, 273, 278
<code>\GTS@@mainColor</code>	99, 100, 101, 102, 108, 110, 112, 114, 127, 134, 242, 243, 244, 245, 251, 253, 255, 257, 270, 277
<code>\GTS@@relation</code>	122, 123, 124, 132, 135, 265, 266, 267, 275, 278
<code>\GTS@@secondColor</code>	103, 104, 105, 106, 108, 110, 112, 114, 128, 134, 246, 247, 248, 249, 251, 253, 255, 257, 271, 277
<code>\GTS@a</code>	4, 34, 94, 96, 141, 142, 143, 147, 148, 149, 153, 154, 155, 159, 160, 161, 167, 168, 169, 173, 174, 175, 320, 321, 322, 323, 328, 329, 330, 331, 337, 339, 340, 341, 349, 351, 352, 353, 361, 363, 364, 365, 373, 375, 376, 377
<code>\GTS@b</code>	4, 92, 134
<code>\GTS@drawDownArrow</code>	16
<code>\GTS@drawSlash</code>	14
<code>\GTS@drawUpArrow</code>	15

\GTS@i	.. 7, 237, 239, 284, 285, 286, 290, 291, 292, 296, 297, 298, 302, 303, 304, 308, 309, 310, 314, 315, 316, 384, 385, 386, 387, 393, 395, 396, 397, 402, 403, 404, 405, 411, 413, 414, 415, 422, 423, 424, 425, 431, 433, 434, 435
\GTS@j	7, 277
\GTS@mainColor	10, 26, 27, 53, 67, 73, 204
\GTS@mainColorBackup	12, 26, 86, 229
\GTS@mainColorTemp	25, 27
\GTS@newcommand	.. 4, <u>18</u> , 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 179, 180, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 339, 340, 341, 342, 345, 346, 347, 348, 351, 352, 353, 354, 357, 358, 359, 360, 363, 364, 365, 366, 369, 370, 371, 372, 375, 376, 377, 378, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 413, 414, 415, 416, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 433, 434, 435, 436, 439, 440, 441, 452, 462, 463, 464, 465, 475, 476, 477, 478
\GTS@newprotectedcommand	4, <u>21</u> , 442, 453, 466
\GTS@secondColor	11, 31, 32, 45, 49, 57, 61, 192, 196, 200, 210, 216
\GTS@secondColorBackup	13, 31, 89, 232
\GTS@secondColorTemp	30, 32
\GTSa	4, <u>141</u> , 144, 145, 146
\GTSadj	4, <u>141</u>
\GTSadjacency	4, <u>141</u>
\GTSadjacencyMonochrome	9
\GTSadjacencyMonochromeRelation	9, <u>323</u> , 324
\GTSadjacencyRelation	5, <u>144</u>
\GTSadjMon	9
\GTSadjMono	9
\GTSadjMonoRel	9, <u>323</u>
\GTSadjMonRel	9, <u>323</u>
\GTSadjRel	5, <u>144</u>
\GTSaM	9, 324, 325, 326, 327
\GTSaMR	9, <u>323</u>
\GTSaR	5, <u>144</u>
\GTSdAD	6, 170, 171, 172
\GTSdADM	11, <u>359</u> , 367, 369, 370, 371
\GTSdADMR	11, <u>365</u>
\GTSdADR	6, <u>169</u>
\GTSdAU	5, <u>153</u> , 156, 157, 158
\GTSdAUM	10, <u>335</u> , 343, 345, 346, 347
\GTSdAUMR	10, <u>341</u>
\GTSdAUR	5, <u>155</u>
\GTSdirAdjDown	6
\GTSdirAdjDownMon	11, <u>359</u>
\GTSdirAdjDownMono	11, <u>359</u>
\GTSdirAdjDownMonoRel	11, <u>365</u>
\GTSdirAdjDownMonRel	11, <u>365</u>
\GTSdirAdjDownRel	6, <u>169</u>
\GTSdirAdjUp	5, <u>153</u>
\GTSdirAdjUpMon	10, <u>335</u>
\GTSdirAdjUpMono	10, <u>335</u>
\GTSdirAdjUpMonoRel	10, <u>341</u>
\GTSdirAdjUpMonRel	10, <u>341</u>
\GTSdirAdjUpRel	5, <u>155</u>
\GTSdirectedAdjacencyDown	6
\GTSdirectedAdjacencyDownMonochrome	11, <u>359</u> , 360
\GTSdirectedAdjacencyDownMonochromeRelation	11, <u>365</u> , 366
\GTSdirectedAdjacencyDownRelation	6, <u>169</u> , 170
\GTSdirectedAdjacencyUp	5, <u>153</u>

\GTSdirectedAdjacencyUpMonochrome	10, <u>335</u> , 336
\GTSdirectedAdjacencyUpMonochromeRelation	10, <u>341</u> , 342
\GTSdirectedAdjacencyUpRelation	5, <u>155</u> , 156
\GTSi	7, 287, 288, 289
\GTSiI	8, 311, 312, 313
\GTSiIM	13, 426, 427, 428, 429
\GTSiIMR	13, <u>425</u>
\GTSiIR	9
\GTSiM	12, 388, 389, 390, 391
\GTSiMR	12, <u>387</u>
\GTSinc	7
\GTSincidence	7
\GTSincidenceMonochrome	12
\GTSincidenceMonochromeRelation	12, <u>387</u> , 388
\GTSincidenceRelation	7
\GTSincMon	12
\GTSincMono	12
\GTSincMonoRel	12, <u>387</u>
\GTSincMonRel	12, <u>387</u>
\GTSincRel	7
\GTSinInc	8
\GTSinIncidence	8
\GTSinIncidenceMonochrome	13
\GTSinIncidenceMonochromeRelation	13, <u>425</u> , 426
\GTSinIncidenceRelation	9
\GTSinIncMon	13
\GTSinIncMono	13
\GTSinIncMonoRel	13, <u>425</u>
\GTSinIncMonRel	13, <u>425</u>
\GTSinIncRel	9
\GTSiR	7
\GTSmA	14, 463, 464, 465
\GTSmAR	14
\GTSmetaAnd	14
\GTSmetaAndRel	14
\GTSmetaAndRelation	14
\GTSmetaNot	14
\GTSmetaOr	15
\GTSmetaOrRel	15
\GTSmetaOrRelation	15
\GTSmN	14
\GTSmO	15, 476, 477, 478
\GTSmOR	15
\GTSnDAD	6, <u>172</u> , 177, 179, 180
\GTSnDADM	11, <u>371</u> , 379, 381, 382, 383
\GTSnDADMR	11, <u>377</u>
\GTSnDADR	6, <u>175</u>
\GTSnDAU	6, <u>158</u> , 163, 165, 166
\GTSnDAUM	10, <u>347</u> , 355, 357, 358, 359
\GTSnDAUMR	11, <u>353</u>
\GTSnDAUR	6, <u>161</u>
\GTSnegatedDirectedAdjacencyDown	6, <u>172</u> , 173
\GTSnegatedDirectedAdjacencyDownMonochrome	11, <u>371</u> , 372
\GTSnegatedDirectedAdjacencyDownMonochromeRelation	11, <u>377</u> , 378
\GTSnegatedDirectedAdjacencyDownRelation	6, <u>175</u> , 176
\GTSnegatedDirectedAdjacencyUp	6, <u>158</u> , 159
\GTSnegatedDirectedAdjacencyUpMonochrome	10, <u>347</u> , 348
\GTSnegatedDirectedAdjacencyUpMonochromeRelation	11, <u>353</u> , 354
\GTSnegatedDirectedAdjacencyUpRelation	6, <u>161</u> , 162
\GTSnegatedIncidence	7
\GTSnegatedIncidenceMonochrome	12, <u>391</u> , 392

\GTSnegatedIncidenceMonochromeRelation	12, <u>397</u> , 398
\GTSnegatedIncidenceRelation	8, <u>292</u> , 293
\GTSnegatedInIncidence	9
\GTSnegatedInIncidenceMonochrome	14, <u>429</u> , 430
\GTSnegatedInIncidenceMonochromeRelation	14, <u>435</u> , 436
\GTSnegatedInIncidenceRelation	9, <u>316</u> , 317
\GTSnegatedOutIncidence	8
\GTSnegatedOutIncidenceMonochrome	13, <u>409</u> , 410
\GTSnegatedOutIncidenceMonochromeRelation	13, <u>415</u> , 416
\GTSnegatedOutIncidenceRelation	8, <u>304</u> , 305
\GTSnegDirAdjDown	6, <u>172</u>
\GTSnegDirAdjDownMon	11, <u>371</u>
\GTSnegDirAdjDownMono	11, <u>371</u>
\GTSnegDirAdjDownMonoRel	11, <u>377</u>
\GTSnegDirAdjDownMonRel	11, <u>377</u>
\GTSnegDirAdjDownRel	6, <u>175</u>
\GTSnegDirAdjUp	6, <u>158</u>
\GTSnegDirAdjUpMon	10, <u>347</u>
\GTSnegDirAdjUpMono	10, <u>347</u>
\GTSnegDirAdjUpMonoRel	11, <u>353</u>
\GTSnegDirAdjUpMonRel	11, <u>353</u>
\GTSnegDirAdjUpRel	6, <u>161</u>
\GTSnegInc	7
\GTSnegIncMon	12, <u>391</u>
\GTSnegIncMono	12, <u>391</u>
\GTSnegIncMonoRel	12, <u>397</u>
\GTSnegIncMonRel	12, <u>397</u>
\GTSnegIncRel	8, <u>292</u>
\GTSnegInInc	9
\GTSnegInIncMon	14, <u>429</u>
\GTSnegInIncMono	14, <u>429</u>
\GTSnegInIncMonoRel	14, <u>435</u>
\GTSnegInIncMonRel	14, <u>435</u>
\GTSnegInIncRel	9, <u>316</u>
\GTSnegOutInc	8
\GTSnegOutIncMon	13, <u>409</u>
\GTSnegOutIncMono	13, <u>409</u>
\GTSnegOutIncMonoRel	13, <u>415</u>
\GTSnegOutIncMonRel	13, <u>415</u>
\GTSnegOutIncRel	8, <u>304</u>
\GTSnI	7, 293, 294, 295
\GTSnII	9, 317, 318, 319
\GTSnIIM	14, <u>429</u> , 437, 439, 440, 441
\GTSnIIMR	14, <u>435</u>
\GTSnIIR	9, <u>316</u>
\GTSnIM	12, <u>391</u> , 398, 399, 400, 401
\GTSnIMR	12, <u>397</u>
\GTSnIR	8, <u>292</u>
\GTSnOI	8, 305, 306, 307
\GTSnOIM	13, <u>409</u> , 417, 419, 420, 421
\GTSnOIMR	13, <u>415</u>
\GTSnOIR	8, <u>304</u>
\GTSoI	8, 299, 300, 301
\GTSoIM	12, 406, 407, 408, 409
\GTSoIMR	13, <u>405</u>
\GTSoIR	8
\GTSoutInc	8
\GTSoutIncidence	8
\GTSoutIncidenceMonochrome	12
\GTSoutIncidenceMonochromeRelation	13, <u>405</u> , 406
\GTSoutIncidenceRelation	8

\GTSoutIncMon	12
\GTSoutIncMono	12
\GTSoutIncMonoRel	13, 405
\GTSoutIncMonRel	13, 405
\GTSoutIncRel	8
\GTSsetMainColor	4, 24, 36, 86, 183, 229
\GTSsetSecondColor	4, 29, 39, 89, 186, 232
\GTSu	5, 147, 150, 151, 152
\GTSuM	9, 332, 333, 334, 335
\GTSuMR	10, 331
\GTSunadj	5, 147
\GTSunadjacency	5, 147
\GTSunadjacencyMonochrome	9
\GTSunadjacencyMonochromeRelation	10, 331, 332
\GTSunadjacencyRelation	5, 150
\GTSunadjMon	9
\GTSunadjMono	9
\GTSunadjMonoRel	10, 331
\GTSunadjMonRel	10, 331
\GTSunadjRel	5, 150
\GTSuR	5, 150
\GTSxa	4, 99
\GTSxadj	4, 99
\GTSxadjacency	4, 99
\GTSxi	7
\GTSxinc	7
\GTSxincidence	7